Susie-relax with the "irrelevance of the null SNPs" property

1 SER-relax-prime

Now we consider another approximation to truly achieve the "irrelevance of the null SNPs" property. It leads to another model that I call 'SER-relax-prime'.

Recall the SER-relax (conditional on $\gamma = 1$):

$$x \mid \beta, \gamma = 1 \ \sim \ \mathcal{N}\left(\beta \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}, \ \frac{1}{N} \begin{pmatrix} 1 & \mu_1^T \\ \mu_1 & \bar{R}_{-1,-1} + N\beta^2\tau^2 S_1 \end{pmatrix}\right) \tag{1}$$

Matthew + Gemini suggest omitting β^2 (replacing β^2 by 1) from the variance in the formula above. Similar to computation in PIP from the previous file, we have

$$\begin{aligned} p(x \mid \beta, \gamma = j) = & \frac{N^{J/2}}{(2\pi)^{J/2}\, |\bar{R}|^{1/2}\, (1 + N\tau^2)^{(J-1)/2}} \exp\left(-\frac{N}{2(1+N\tau^2)} q\right) \\ & \times \exp\left(-\frac{N^2\tau^2}{2(1+N\tau^2)} x_j^2\right) \exp\left(N\beta x_j - \frac{N}{2}\beta^2\right). \end{aligned} \tag{2}$$

Integrating out $\beta \sim \mathcal{N}(0, \sigma_0^2)$, we have

$$\begin{aligned} p(x \mid \gamma = j) = & \frac{N^{J/2}}{(2\pi)^{J/2}\, |\bar{R}|^{1/2}\, (1 + N\tau^2)^{(J-1)/2}} \exp\left(-\frac{N}{2(1+N\tau^2)} q\right) \\ & \times \sqrt{\frac{1}{1+N\sigma_0^2}} \times \exp\left(\left(\frac{N\sigma_0^2}{1+N\sigma_0^2} - \frac{N\tau^2}{1+N\tau^2}\right) \frac{N x_j^2}{2}\right). \end{aligned} \tag{3}$$

Leading to

$$p(\gamma = j | x) \propto \exp\left(\left(\frac{N\sigma_0^2}{1+N\sigma_0^2} - \frac{N\tau^2}{1+N\tau^2}\right) \frac{N x_j^2}{2}\right). \tag{4}$$

So we truly have the desired property. However, something is not right about the formula above: The scalar in the exponential can turn into negative if $\sigma_0^2 < \tau^2$ (so that a smaller Z-score leads to a bigger PIP?). It is because we are replacing β^2 by 1 in the variance of the likelihood of $p(x|\beta, \gamma = j)$. Maybe the approximation $\beta^2 \approx \sigma_0^2$ makes more sense.

SER-relax-prime. This model approximates β^2 in the variance of (1) by σ_0^2. It leads to

$$x \mid \beta, \gamma = 1 \;\sim\; \mathcal{N}\left(\beta \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}, \; \frac{1}{N} \begin{pmatrix} 1 & \mu_1^T \\ \mu_1 & \bar{R}_{-1,-1} + N\sigma_0^2 \tau^2 S_1 \end{pmatrix}\right), \tag{5}$$

and similarly for other $\gamma = j$ (recall $\mu_1 = \bar{R}_{-1,1}$). The PIP in this model is:

$$p(\gamma = j|x) \propto \exp\left(\left(\frac{N\sigma_0^2}{1 + N\sigma_0^2} - \frac{N\sigma_0^2\tau^2}{1 + N\sigma_0^2\tau^2}\right)\frac{Nx_j^2}{2}\right). \tag{6}$$

Note that $\tau^2 = 1/(\nu_0 + 3) \leq 1/3 < 1$, so that $N\sigma_0^2\tau^2 < N\sigma_0^2$, implying

$$\frac{N\sigma_0^2}{1 + N\sigma_0^2} - \frac{N\sigma_0^2\tau^2}{1 + N\sigma_0^2\tau^2} > 0.$$

This quantity tends to $\dfrac{N\sigma_0^2}{1 + N\sigma_0^2}$ as $\tau^2 \to 0$ (recovering susie-rss). One interpretation of (6) is that PIP might be more a little more diffuse when we are more uncertain about the in-sample R (which can be a bit funny when we think about it, because if $L = 1$ the LD does not matter and the PIP ideally should be similar to susie-rss. But I guess nothing is perfect, and this really shows how Susie-relax does "relax" the signal.)

However, one can show that this PIP retains some important properties:

1. PIP of SNP j is exponentially proportional to the Z-score square
2. if two SNPs have similar Z-score, they will have the same PIP
3. To compare this PIP with susie-rss, write

   $$t = N\sigma_0^2, \qquad s = \tau^2.$$

   The susie-rss coefficient in the exponent is

   $$a_{\text{rss}} = \frac{t}{1+t},$$

   whereas the SER-relax-prime coefficient is

   $$a_{\text{prime}} = \frac{t}{1+t} - \frac{ts}{1+ts}.$$

   Thus

   $$0 \leq a_{\text{rss}} - a_{\text{prime}} = \frac{ts}{1+ts} \leq ts.$$

   If $\nu_0 \geq 10$, then $s = \tau^2 = 1/(\nu_0 + 3) \leq 1/13$, so

   $$a_{\text{rss}} - a_{\text{prime}} \leq \frac{t}{13+t} \leq \frac{t}{13}.$$

   Equivalently, for any two SNPs j and k,

   $$\left|\log \frac{p_{\text{prime}}(\gamma = j \mid x)}{p_{\text{prime}}(\gamma = k \mid x)} - \log \frac{p_{\text{rss}}(\gamma = j \mid x)}{p_{\text{rss}}(\gamma = k \mid x)}\right| = \frac{N}{2}\frac{ts}{1+ts}|x_j^2 - x_k^2| \leq \frac{N}{2}\frac{t}{13+t}|x_j^2 - x_k^2|.$$

   Hence, when $t = N\sigma_0^2$ is not too large, $\nu_0 = 10$ already gives only a small flattening of the susie-rss log-odds. The approximation becomes exact as $\nu_0 \to \infty$.

Posterior distribution of β under SER-relax-prime. Under (5), the covariance no longer depends on β. Therefore, conditional on $\gamma = j$, the likelihood as a function of β is

$$p(x \mid \beta, \gamma = j) \propto \exp\left(N\beta x_j - \frac{N}{2}\beta^2\right),$$

where all remaining terms are constant in β. Combining this likelihood with the prior $\beta \sim \mathcal{N}(0, \sigma_0^2)$ gives

$$p(\beta \mid x, \gamma = j) = \mathcal{N}(m_j, v), \tag{7}$$

$$v = \left(N + \sigma_0^{-2}\right)^{-1}, \tag{8}$$

$$m_j = vNx_j. \tag{9}$$

Thus the conditional posterior variance is shared across coordinates, and the posterior mean depends on the data only through the active marginal statistic x_j. In particular, $p(\beta \mid x, \gamma = j)$ does not depend on $q = x^T \bar{R}^{-1} x$.

2 susie-relax-prime

How to extend (5) to the sum of a single-effect type of model? Ideally, each of the single-effect models should have the same noise model (like $\mathcal{N}(0, \bar{R}/N)$) to enable IBSS-like inference. I think it is cleanest to introduce a latent variable η_ℓ and an effect contribution vector θ_ℓ for each single-effect component. For $\ell = 1, \ldots, L$,

$$\gamma_\ell \sim \text{Unif}\{1, \ldots, J\}, \qquad \beta_\ell \sim \mathcal{N}(0, \sigma_{0\ell}^2).$$

Conditional on $\gamma_\ell = j$, define

$$\theta_\ell = \beta_\ell \bar{R}_{\cdot j} + \eta_{\ell j},$$

where $\eta_{\ell j, j} = 0$ and

$$\eta_{\ell j, -j} \sim \mathcal{N}(0, \sigma_{0\ell}^2 \tau^2 S_j).$$

The observation model is

$$x \mid \theta_1, \ldots, \theta_L \sim \mathcal{N}\left(\sum_{\ell=1}^{L} \theta_\ell, \frac{\bar{R}}{N}\right).$$

This is a proper generative model. If we integrate out $\eta_{\ell j}$ for a single effect, we recover the SER-relax-prime covariance in (5), with σ_0^2 replaced by $\sigma_{0\ell}^2$.

Variational family and ELBO. Use the mean-field approximation

$$q = \prod_{\ell=1}^{L} q_\ell(\gamma_\ell, \beta_\ell, \eta_\ell),$$

with

$$q_\ell(\gamma_\ell = j, \beta_\ell, \eta_{\ell j}) = \alpha_{\ell j} q_{\ell j}(\beta_\ell) q_{\ell j}(\eta_{\ell j}).$$

The ELBO is the standard variational objective

$$\mathcal{L}(q) = E_q \log p(x \mid \theta_1, \dots, \theta_L) + \sum_{\ell=1}^{L} E_q \log p(\gamma_\ell, \beta_\ell, \eta_\ell \mid \gamma_\ell) - \sum_{\ell=1}^{L} E_q \log q_\ell(\gamma_\ell, \beta_\ell, \eta_\ell). \tag{10}$$

Holding all factors except q_ℓ fixed, the optimal coordinate update is

$$\log q_\ell^*(\gamma_\ell, \beta_\ell, \eta_\ell) = E_{-\ell} \log p(x, \{\gamma_k, \beta_k, \eta_k\}_{k=1}^{L}) + \text{const}.$$

Let

$$\bar{\theta}_k = E_q(\theta_k), \qquad r_\ell = x - \sum_{k \neq \ell} \bar{\theta}_k.$$

Then the q_ℓ update is exactly the single-effect SER-relax-prime posterior computed with x replaced by r_ℓ.

Coordinate posterior. For each effect define

$$\kappa_\ell = \frac{N\sigma_{0\ell}^2}{1 + N\sigma_{0\ell}^2}, \qquad \rho_\ell = \frac{N\sigma_{0\ell}^2\tau^2}{1 + N\sigma_{0\ell}^2\tau^2}.$$

The coordinate weights are

$$\alpha_{\ell j} = \frac{\exp\left\{\frac{N}{2}(\kappa_\ell - \rho_\ell) r_{\ell j}^2\right\}}{\sum_{k=1}^{J} \exp\left\{\frac{N}{2}(\kappa_\ell - \rho_\ell) r_{\ell k}^2\right\}}. \tag{11}$$

The common factor involving $q_\ell^{\text{quad}} = r_\ell^T \bar{R}^{-1} r_\ell$ cancels in this normalization. Conditional on $\gamma_\ell = j$,

$$q_{\ell j}(\beta_\ell) = \mathcal{N}(m_{\ell j}, v_\ell), \tag{12}$$

$$v_\ell = \left(N + \sigma_{0\ell}^{-2}\right)^{-1}, \tag{13}$$

$$m_{\ell j} = v_\ell N r_{\ell j} = \kappa_\ell r_{\ell j}. \tag{14}$$

The relaxed part also has a Gaussian posterior. Its nonzero block is

$$q_{\ell j}(\eta_{\ell j,-j}) = \mathcal{N}(\bar{\eta}_{\ell j,-j}, W_{\ell j}), \tag{15}$$

$$\bar{\eta}_{\ell j,-j} = \rho_\ell (r_{\ell,-j} - \mu_j r_{\ell j}), \tag{16}$$

$$W_{\ell j} = \frac{\sigma_{0\ell}^2 \tau^2}{1 + N\sigma_{0\ell}^2\tau^2} S_j. \tag{17}$$

Equivalently, as a J-vector,

$$\bar{\eta}_{\ell j} = \rho_\ell (r_\ell - r_{\ell j} \bar{R}_{\cdot j}).$$

Posterior mean contribution. The posterior mean contribution of effect ℓ must include both the posterior mean of $\beta_\ell \bar{R}_{\cdot j}$ and the posterior mean of the relaxed component $\eta_{\ell j}$:

$$\begin{aligned}\bar{\theta}_\ell &= \sum_{j=1}^{J} \alpha_{\ell j} \left[m_{\ell j} \bar{R}_{\cdot j} + \rho_\ell (r_\ell - r_{\ell j} \bar{R}_{\cdot j}) \right] \\ &= \bar{R}(\alpha_\ell \odot m_\ell) + \rho_\ell r_\ell - \rho_\ell \bar{R}(\alpha_\ell \odot r_\ell). \end{aligned} \tag{18}$$

This correction is what makes the residual update consistent with the variational posterior under the latent contribution model.

ELBO computation. Let $\Omega = \bar{R}^{-1}$ and

$$C_x = -\frac{J}{2}\log(2\pi) - \frac{1}{2}\log\left|\frac{\bar{R}}{N}\right|.$$

For each (ℓ, j) define

$$c_\ell = \frac{\sigma_{0\ell}^2 \tau^2}{1 + N\sigma_{0\ell}^2 \tau^2}, \qquad C_{\ell j} = (r_{\ell,-j} - \mu_j r_{\ell j})^T S_j^{-1} (r_{\ell,-j} - \mu_j r_{\ell j}).$$

Using the block inverse identity,

$$C_{\ell j} = r_\ell^T \Omega r_\ell - r_{\ell j}^2,$$

so all $C_{\ell j}$ can be computed from the scalar $r_\ell^T \Omega r_\ell$ and the coordinate-wise residuals. The posterior second moments are

$$B_{\ell j} = E_q(\beta_\ell^2 \mid \gamma_\ell = j) = m_{\ell j}^2 + v_\ell, \tag{19}$$

$$D_{\ell j} = E_q(\eta_{\ell j,-j}^T S_j^{-1} \eta_{\ell j,-j}) = \rho_\ell^2 C_{\ell j} + (J-1)c_\ell. \tag{20}$$

Because $\bar{R}_{\cdot j}^T \Omega = e_j^T$ and $\eta_{\ell j,j} = 0$, the cross term between $\beta_\ell \bar{R}_{\cdot j}$ and $\eta_{\ell j}$ is zero. Hence

$$E_q(\theta_\ell^T \Omega \theta_\ell) = \sum_{j=1}^{J} \alpha_{\ell j}(B_{\ell j} + D_{\ell j}).$$

Let $\bar{\theta} = \sum_{\ell=1}^{L} \bar{\theta}_\ell$. A fast way to compute the expected quadratic loss is

$$E_q\left[(x - \sum_{\ell=1}^{L}\theta_\ell)^T \Omega (x - \sum_{\ell=1}^{L}\theta_\ell)\right] = (x-\bar{\theta})^T \Omega (x - \bar{\theta}) + \sum_{\ell=1}^{L} V_\ell,$$

where

$$V_\ell = \sum_{j=1}^{J} \alpha_{\ell j}(B_{\ell j} + D_{\ell j}) - \bar{\theta}_\ell^T \Omega \bar{\theta}_\ell.$$

Thus the expected log-likelihood is

$$E_q \log p(x \mid \theta_1, \ldots, \theta_L) = C_x - \frac{N}{2}\left[(x-\bar{\theta})^T \Omega (x-\bar{\theta}) + \sum_{\ell=1}^{L} V_\ell\right].$$

The remaining terms are most compactly written as KL divergences. Define

$$\mathrm{KL}^{\beta}_{\ell j} = \frac{1}{2}\left[\log\frac{\sigma^2_{0\ell}}{v_\ell} + \frac{B_{\ell j}}{\sigma^2_{0\ell}} - 1\right], \tag{21}$$

and

$$\begin{aligned}\mathrm{KL}^{\eta}_{\ell j} &= \frac{1}{2}\left[(J-1)\log\frac{\sigma^2_{0\ell}\tau^2}{c_\ell} + \frac{D_{\ell j}}{\sigma^2_{0\ell}\tau^2} - (J-1)\right] \\ &= \frac{1}{2}\left[(J-1)\log(1+N\sigma^2_{0\ell}\tau^2) + \frac{D_{\ell j}}{\sigma^2_{0\ell}\tau^2} - (J-1)\right].\end{aligned} \tag{22}$$

The determinants $|S_j|$ cancel in $\mathrm{KL}^{\eta}_{\ell j}$ because $W_{\ell j} = c_\ell S_j$. The full ELBO is therefore

$$\begin{aligned}\mathcal{L}(q) = C_x &- \frac{N}{2}\left[(x-\bar\theta)^T\Omega(x-\bar\theta) + \sum_{\ell=1}^{L} V_\ell\right] \\ &- \sum_{\ell=1}^{L}\sum_{j=1}^{J}\alpha_{\ell j}\left[\log(J\alpha_{\ell j}) + \mathrm{KL}^{\beta}_{\ell j} + \mathrm{KL}^{\eta}_{\ell j}\right].\end{aligned} \tag{23}$$

This is the expression to use in code. It only requires:

1. the fitted mean $\bar\theta$ and one quadratic $(x-\bar\theta)^T\Omega(x-\bar\theta)$;
2. for each effect, $\bar\theta_\ell^T\Omega\bar\theta_\ell$;
3. for each effect, $r_\ell^T\Omega r_\ell$, which gives all $C_{\ell j} = r_\ell^T\Omega r_\ell - r^2_{\ell j}$;
4. coordinate-wise scalar operations for $B_{\ell j}$, $D_{\ell j}$, and the KL terms.

No determinant of S_j is needed, and no dense $W_{\ell j}$ matrix has to be formed.

Variational empirical Bayes updates. Holding the variational distribution fixed, maximize the ELBO over $\sigma^2_{0\ell}$ and τ^2. Define the effect-level averages

$$B_\ell = \sum_{j=1}^{J}\alpha_{\ell j}B_{\ell j}, \qquad D_\ell = \sum_{j=1}^{J}\alpha_{\ell j}D_{\ell j}.$$

The part of the ELBO that depends on $\sigma^2_{0\ell}$ is

$$-\frac{J}{2}\log\sigma^2_{0\ell} - \frac{1}{2\sigma^2_{0\ell}}\left(B_\ell + \frac{D_\ell}{\tau^2}\right),$$

so the closed-form VEB update is

$$\hat\sigma^2_{0\ell} = \frac{1}{J}\left(B_\ell + \frac{D_\ell}{\tau^2}\right).$$

This is the average posterior second moment of the latent effect: one degree of freedom comes from β_ℓ, and $J-1$ degrees of freedom come from the relaxed LD-column perturbation after rescaling by τ^2.

Holding $\{\sigma^2_{0\ell}\}_{\ell=1}^L$ fixed, the τ^2-dependent part of the ELBO is

$$-\frac{L(J-1)}{2}\log\tau^2 - \frac{1}{2\tau^2}\sum_{\ell=1}^{L}\frac{D_\ell}{\sigma^2_{0\ell}},$$

so

$$\hat{\tau}^2 = \frac{1}{L(J-1)}\sum_{\ell=1}^{L}\frac{D_\ell}{\sigma^2_{0\ell}}.$$

This update estimates the average standardized size of the relaxed LD-column perturbations. A small value means the posterior does not need much LD relaxation and the method approaches susie-rss; a large value means the data are being explained partly through LD-column uncertainty. In practice these EB updates can be alternated with the variational updates and bounded to avoid degenerate values.

IBSS-like coordinate ascent. The resulting algorithm is coordinate ascent on $\mathcal{L}(q)$:

1. Initialize $\alpha_{\ell j}$, $m_{\ell j}$, v_ℓ, $\bar{\eta}_{\ell j}$, and $\bar{\theta}_\ell$ for $\ell = 1,\dots,L$.
2. For each effect ℓ, form
$$r_\ell = x - \sum_{k\neq\ell}\bar{\theta}_k.$$
3. Compute $v_\ell = (N+\sigma_{0\ell}^{-2})^{-1}$, $m_{\ell j} = v_\ell N r_{\ell j}$, and $\bar{\eta}_{\ell j} = \rho_\ell(r_\ell - r_{\ell j}\bar{R}_{\cdot j})$ for all j.
4. Update the coordinate weights by
$$\alpha_{\ell j} \propto \exp\left\{\frac{N}{2}(\kappa_\ell - \rho_\ell)r_{\ell j}^2\right\}.$$
5. Recompute
$$\bar{\theta}_\ell = \bar{R}(\alpha_\ell \odot m_\ell) + \rho_\ell r_\ell - \rho_\ell \bar{R}(\alpha_\ell \odot r_\ell).$$
6. Optionally update $\sigma^2_{0\ell}$ and τ^2 using the VEB formulas, then refresh κ_ℓ, ρ_ℓ, v_ℓ, and $W_{\ell j}$.
7. Sweep over $\ell = 1,\dots,L$ until the ELBO or fitted mean $\sum_{\ell=1}^L \bar{\theta}_\ell$ stabilizes.

Comment on the robustness. We knew from the experiments that under LD mismatch, Susie-RSS's signal $R\bar{b}$ can't fit exactly to the data x (previously denoted by v) and therefore needs more L. This model allow for some deviation from $R\bar{b}$ by additionally using $(\eta_\ell)_{\ell=1}^L$. Looking at the final ELBO (23), the choice of L now is a battle between τ^2 and $\sigma^2_{0\ell}$ in equations (21) and (22): If introduce a little more τ^2 can save us the cost of one more $\sigma^2_{0\ell}$, the model will rather set τ^2 to slightly larger (allowing more deviation) and set that extra $\sigma^2_{0\ell}$ to 0, leading to more conservative behavior.